CHEAT SHEET

1. Defining and calling functions

- A function is a reusable block of code that perform a task.

Ex.
```
def greet():
    print("Hello")
greet()
```

2. Print and return

- print() displays something on the screen.
- return Sends a value back to the code that called the function.

```
def add(a, b):
    return a+b
result = add(2, 3)
print(result) # 5
```

3. Arguments

Arguments are values passed into a function.

```
def greet(name, age=18):
    print(name, age)
greet("John")
greet(name="Mary", age=20)
```

4. *args and **kwargs

- *args accepts any number of positional arguments
- **kwargs accepts any number of keywords arguments

```
def show(*args, **kwargs):
    print(args)
    print(kwargs)
show(1, 2, name="John", age=18)
```

5. Scope

Scope determins where the variable can be accessed. A variable created inside a function is usually local to that function

```
def calculate():
    x = 10
    return x
print(calculate()) #1
```

6. Docstring

- A docstring describes what a fuction does. It is written immediately after the function definition.

```
def square(n)
    """Return the square of a number."""
    return n*n
print(square(4)) #16
```

7. Modules

A module is a file or library containing reusable python code

E.g
```
import math
from math import sqrt
import random
print(math.pi)
print(sqrt(25)
print(random.randit(1, 10))
```

8. If __name__ == "__main__":

- This condition makes code run only when the python file is executed directly, not when it is imported as a module.

```
def greet():
    print("Hello!")
if __name__ == "__main__":
    greet()
```

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